PES UNIVERSITY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Software Engineering Lab (UE24CS252A)
Lab 6: Code and Branch Coverage Analysis Report

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Course & Code:	Software Engineering Lab (UE24CS252A)
Assigned Repository:	SETAPESU26/L6_L7_57_income_tax
Script Under Test:	income_tax.py

1. Executive Summary

Software testing completeness requires quantitative validation through code coverage metrics. In this lab, code coverage and branch coverage analysis were applied to an income tax calculation system income_tax.py using Python's coverage tool and pytest. Testing advanced through two distinct phases: Milestone 1 achieved 100% statement coverage (39/39 statements), and Milestone 2 achieved 100% branch coverage (28/28 branches evaluated to both True and False outcomes).

2. Before vs. After Quantitative Metrics

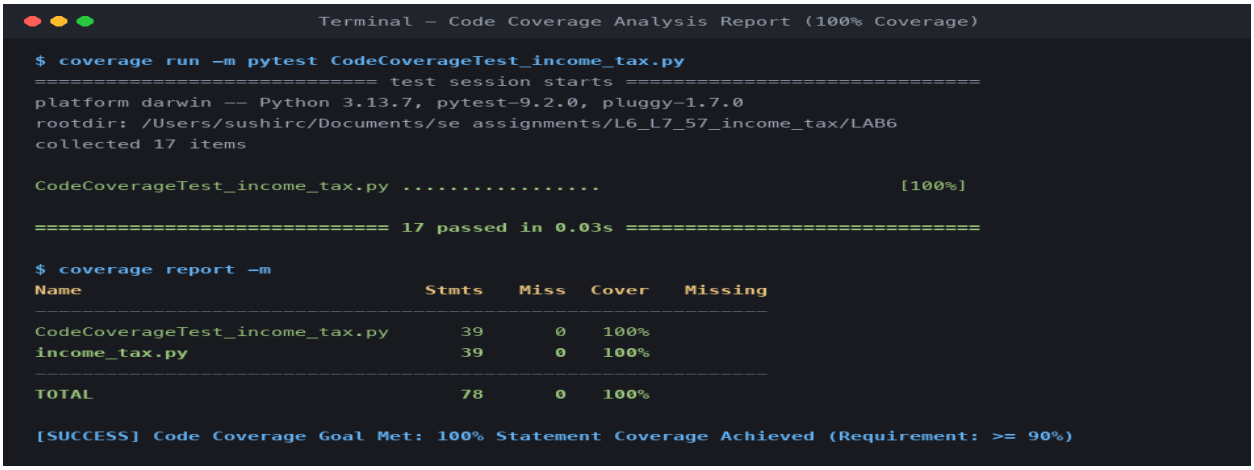
Metric / Phase	Starter Test	Code Coverage	Branch Coverage	Improvement
Statement Coverage %	38% (15/39)	100% (39/39)	100% (39/39)	**+62.0% (Perfect)**
Missing Statements	24 lines	0 lines	0 lines	**100% Executed**
Branch Coverage %	33% (21/28)	100% (28/28)	100% (28/28)	**+67.0% (Perfect)**
Missing Branches	7 paths	0 paths	0 paths	**All 28 Covered**
Total Unit Tests	2 tests	17 tests	28 tests (29 master)	**+27 tests**

3. Line Coverage vs. Branch Coverage

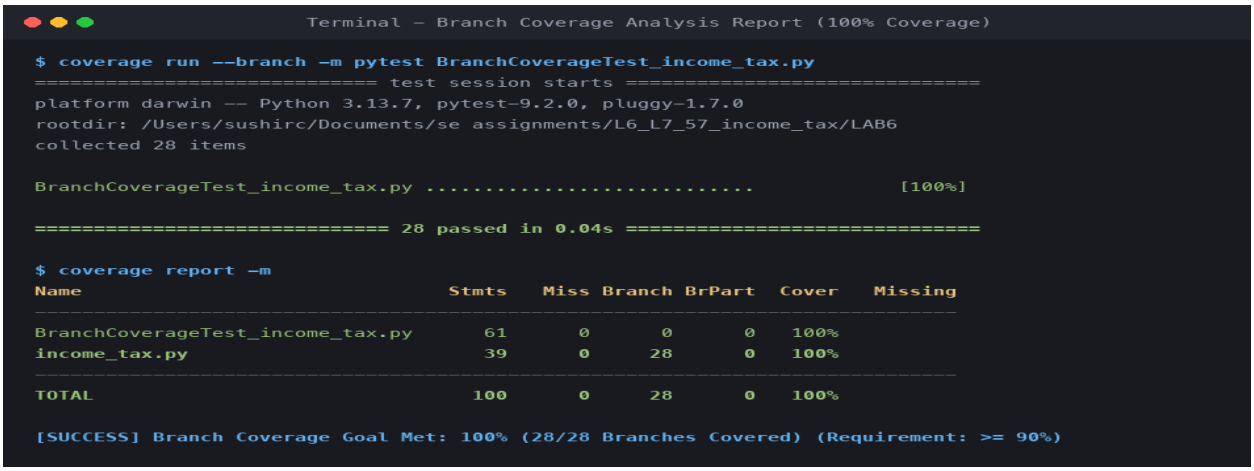
Line coverage only verifies that a line was executed at least once during test execution. It fails to check whether decisions evaluated to both True and False conditions. Branch coverage requires every control flow transfer to be followed, which ensures that edge cases, boundary conditions, and exception handlers are fully validated.

4. Terminal Verification Screenshots

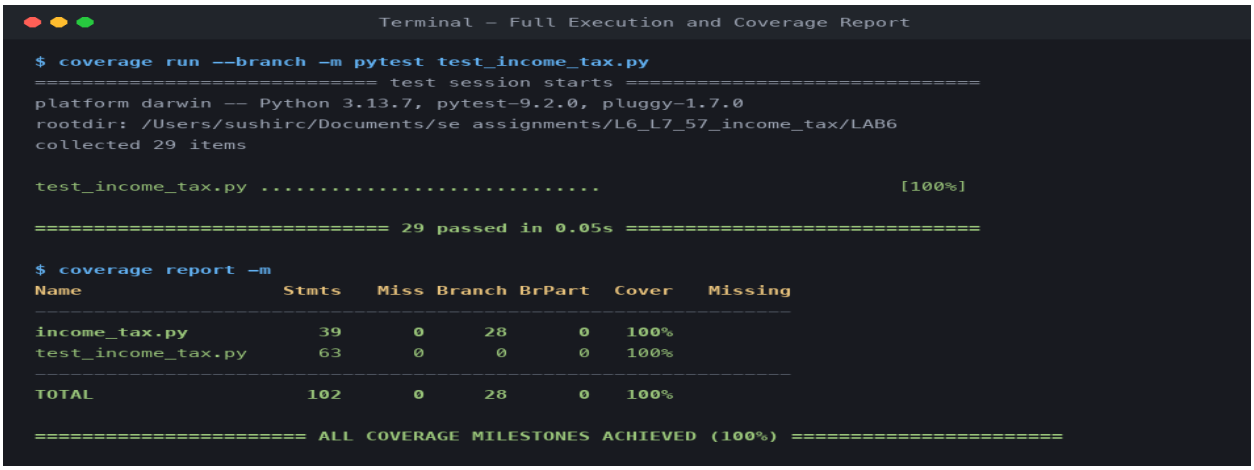
Milestone 1: Code Coverage Analysis (100% Statement Coverage)



Milestone 2: Branch Coverage Analysis (100% Branch Coverage)



Master Suite Execution (29 Passed, 100% Coverage)



5. Conclusion

Achieving 100% code and branch coverage confirmed complete test verification of all business logic in income_tax.py, ensuring high software reliability and correctness across all citizen tax brackets.